# General Mathematics — Exercise Sheet 31

*Taylor series and termwise differentiation*

**Variant: Statements**

**Exercise 1 (A quotient whose missing value has all derivatives).** For $x \neq 0$, let $f(x) = (e^x - 1 - x)/x^2$. Choose $f(0)$ to make $f$ continuous. Use Lecture 31's exponential series to obtain a power-series representation valid for every real $x$, prove that the extension is analytic at zero and $C^\infty$ on $\mathbb{R}$, and find $f^{(k)}(0)$ for every integer $k \geq 0$. Here $f^{(0)} = f$. Justify the extension before differentiating at zero.

**Exercise 2 (A weighted exponential sum).** Find the sum of the power series

$$S(x) = \sum_{n=0}^{\infty} \frac{(n+1)^2 x^n}{n!} \qquad (x \in \mathbb{R}),$$

and evaluate $S(-1)$. Use the derivative theorem and exponential series of Lecture 31. Establish the radius before using the derivative rule.

**Exercise 3 (The binomial sketch and an inverse hyperbolic series).** For $\alpha \in \mathbb{R}$, use the following *binomial expansion*, recorded with sketch status [S]:

$$(1+u)^\alpha = \sum_{n=0}^{\infty} c_n u^n \qquad (|u| < 1).$$

Here $c_0 = 1$, and the remaining coefficients are

$$c_n = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} \qquad (n \geq 1).$$

Its sketch is as follows. For a nonterminating series, the coefficient ratio gives radius 1; for a nonnegative integer $\alpha$, the series terminates. The recurrence $(n+1)c_{n+1} = (\alpha - n)c_n$ gives $(1+u)B' = \alpha B$ for its sum $B$. The derivative of $B(u)(1+u)^{-\alpha}$ is zero, and its value at zero is 1. The constant-function consequence of Lecture 9 then identifies $B$. This is the announced sketch (Stewart, §11.10), using Lecture 31's derivative rule.

Apply this expansion with $\alpha = -1/2$ and integrate to obtain

$$\ln\bigl(x + \sqrt{1+x^2}\bigr) = \sum_{n=0}^{\infty} \frac{(-1)^n p_n x^{2n+1}}{2n+1}, \qquad p_0 = 1, \quad p_n = \prod_{j=1}^{n} \frac{2j-1}{2j}.$$

Prove validity for $-1 \leq x \leq 1$, including uniform convergence at the endpoints, and identify the resulting sum at $x = 1$. The primitive $\ln(x + \sqrt{1+x^2})$ has derivative $1/\sqrt{1+x^2}$ and value zero at zero, as proved by Sheet 24's Euler substitution. The derived representation inherits the supplied sketch status [S].

**Exercise 4 (Four certified decimal places for a Gaussian integral).** Let $I = \int_0^1 e^{-x^2}\,\mathrm{d}x$. Use a finite Taylor polynomial and Lecture 11's signed Lagrange remainder to prove that $I$ rounds to 0.7468 to four decimal places. Use the degree-6 polynomial of $e^u$ at $u = 0$, then put $u = -x^2$. Give an exact rational interval for $I$ and integrate the finite error bound using Lecture 26. No value of the improper Gaussian integral is assumed.

**Exercise 5 (A faster positive series for the logarithm).** Derive, for $|x| < 1$,

$$\ln \frac{1+x}{1-x} = 2\sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}.$$

Use $x = 1/3$ and the first six terms to give a rational approximation to $\ln 2$ with error less than $1/1000000$, including the sign of the error. Use Lecture 31's logarithmic expansion and Lecture 29's convergent-series linearity, grouping and geometric tail.

**Exercise 6 (A complex parameter and its removable value).** For $\theta \in \mathbb{R}$, consider

$$A(\theta) = \sum_{n=0}^{\infty} \frac{(\mathrm{i}\theta)^n}{(n+1)!},$$

where the constant term is 1, including when $\theta = 0$. Prove absolute convergence at every real parameter and uniform convergence on every $[-T, T]$ with $T > 0$. For complex functions, uniform convergence uses the complex modulus in Lecture 30's error definition. Find the real and imaginary parts for $\theta \neq 0$ and the value at zero. Check that the nonzero-parameter formula has the continuous extension you found. Use Lecture 31's imaginary exponential series, Lecture 30's majorant argument and the sine and cosine limits of Lectures 5 and 6.